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Evaluating the CO_2 Solubility Trapping Potential and Geochemical Reactions in Saline Aquifers Under Geological Carbon Sequestration Conditions

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Abstract

CO_2 solubility trapping and CO_2-brine-mineral interactions are critical in evaluating the reliability of CO_2 geological storage. Primarily, a high-temperature and high-pressure reactor was developed to measure the amount of dissolved CO_2 in brine and evaluate the CO_2-brine-mineral interactions under subsurface temperature, pressure, and salinity conditions. The experimental results were subsequently validated by published data in the literature. However, measuring these properties can be time-consuming and resource-intensive. CO_2SolTool, developed by coupling MATLAB with Phreeqc, provides an alternative approach for reliably calculating the CO_2 solubility, properties of CO_2-saturated brine, and CO_2-brine-mineral interactions for carbon geological sequestration. The calculated data for CO_2 solubility, CO_2-brine-mineral interactions, and properties of CO_2-saturated brine are consistent with the experimental findings across a wide range of temperature, pressure, and brine salinity, in the presence or absence of gas impurities.

It I found that the solubility of H_2S is higher than that of CO_2, while the solubility of N_2 and CH_4 is smaller than that of CO_2. The electrolytes, especially the HCO_3^-, have a significant effect on the pH of CO_2-saturated brine. The presence of impurities such as H2S may moderately decrease the pH of the CO_2-saturated brine, as it has higher solubility in brine than CO_2 under the same conditions. The presence of minerals, e.g., calcite and dolomite, may increase the pH level after CO_2-brine-mineral equilibrium. The pH of CO_2-saturated brine is not significantly affected by the presence of anhydrite and gypsum. The zeta potential of calcite is largely influenced by the potential-determining ions, HCO_3^-, CO_3^{2-}, Ca^{2+}, etc. Predicting CO_2 solubility and CO_2-brine-mineral interactions allows for better modeling of the fate and transport of CO_2 within the subsurface, optimizing storage strategies, and predicting the long-term behavior of CO_2 in geological formations.

Keywords: CO_2 Solubility, Phreeqc and MATLAB, Solubility and Mineralization Trapping, CO_2-Brine-Mineral Geochemistry

Introduction

CO_2 solubility trapping refers to the dissolution of CO_2 into the formation brine. Solubility of CO_2 directly affects storage capacity by determining how much CO_2 can be dissolved in formation waters

(Jeon, 2018; Zhang, 2024). Dissolved CO_2 is less likely to migrate out of the storage site compared to free CO_2, thus reducing the potential for leakage through faults or fractures in the caprock (Wei, 2011). It is a key mechanism that substantially enhances the long-term security, effectiveness, and economic feasibility of carbon geological storage (CGS) (Burnside, 2014). After CO_2 is solubilized into brine, the brine composition, pH, and density of the CO_2-saturated brine are also altered (Cui, 2018). This information can be crucial for calculating the migration behavior during long-term CGS.

The information of CO_2 solubility can be obtained by experimental measurements, empirical and theoretical approaches (Wei, 2011). The experimental measurements involve direct measurement of CO_2 solubility under controlled laboratory conditions, typically using high-pressure reactors to simulate subsurface temperature, pressure and salinity conditions, which can be costly and time-consuming. The empirical correlations for CO_2 solubility calculations typically rely on fitting experimental data without necessarily accounting for the detailed thermodynamic interactions, phase behavior, physiochemistry, geochemistry, and molecular mechanisms. Recently, estimating CO_2 solubility in brine using the machine learning method has also attached great interest, which is also heavily relied on the accuracy and sufficiency of available data (Wei et al, 2023).

The dissolution of CO_2 into brine alters the chemical equilibrium within the reservoir, potentially leading to the dissolution or precipitation of minerals that can further influence the caprock integrity and the long-term geochemical interactions between the injected CO_2, reservoir rocks, and formation fluids (Zhang et al, 2019). These interactions can also alter the porosity and permeability of the storage reservoir, impacting the injectivity, caprock integrity, efficiency, capacity, and safety of CGS (Cui, 2018). Therefore, reliably estimating the CO_2 solubility in brine, property of CO_2 saturated brine, and CO_2-brine-mineral interactions are essential for constructing accurate geochemical models, to accurately predict the fate of CO_2 in the subsurface, assess risks, evaluate the structural integrity, and assess the storage efficiencies of geological storage sites (Bui, 2018; Carroll, 2017).

The experimental measurements involve direct measurement of CO_2 solubility under controlled laboratory conditions, typically using high-pressure reactors to simulate subsurface temperature, pressure, and salinity conditions, which can be time-consuming and resource-intensive. Advancing the accuracy of predicting the CO_2 solubility in brine and CO_2-brine-mineral interactions through computational models has attracted great interest. In this study, a high-fidelity setup was primarily developed to measure the amount of dissolved CO_2 in brine and evaluate the CO_2-brine-mineral interactions. Phreeqc, a benchmark software for thermodynamic and geochemistry studies, utilizes equation of state (EOS) and Henry law to predict solubility by considering the chemical potentials and phase equilibria of CO_2 and brine. In this study, the CO_2 solubility results calculated by an in-house developed toolkit CO_2SolTool were compared with the experimental data published in the literature, at wide ranges of temperature, pressure, and brine salinity.

Instruments and Methods

The purity of CO_2 used in this work is higher than 99.995%. Electrolytes with ACS grade was purchased from Sigma-Aldrich. Water was deionized and degassed before the experiment. The lab experiments were performed to measure the CO_2 solubility and evaluate the CO_2-brine-mineral interactions. The experimental setup for measuring the CO_2 solubility and CO_2-brine-mineral interactions is schematically shown in Figure 1.

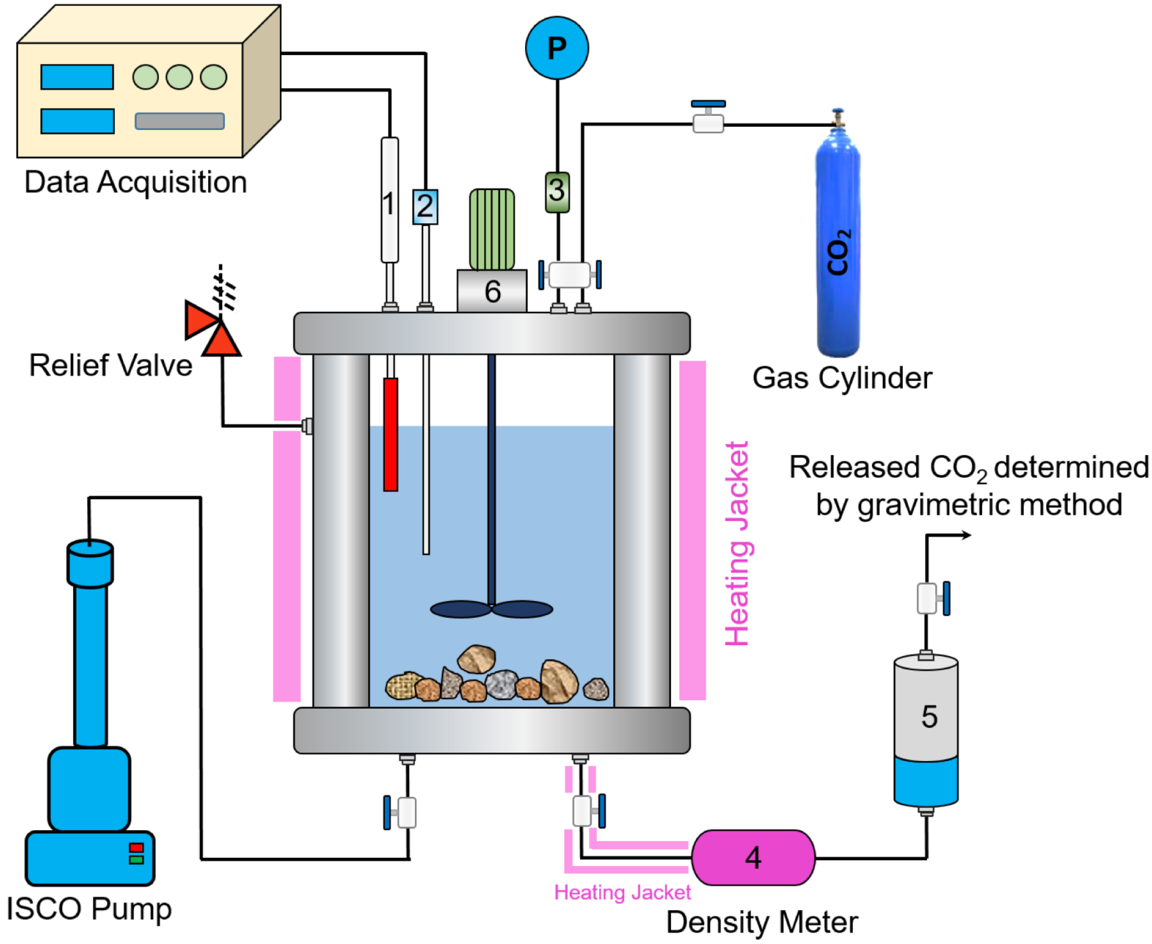


Figure 1—Schematic of the setup for solubility and CO_2-brine-mineral interactions. 1- temperature probe, 2- pH probe, 3- pressure transducer, 4- density meter, 5- sampling cylinder, 6- stirrer.

The experimental setup was adapted from an autoclave produced by Büchiglasuster. The hastelloy cell, known for its corrosion-resistant properties, has a capacity of approximately 500 cm^3 and can withstand conditions up to 500 bar and 300°C. Key components include an ISCO displacement pump, a density meter (Anton Paar DMA HPM), and a high-pressure separator serving as the sampling cylinder. The density meter, placed within a heating jacket, operates within a range of up to 200°C and 1400 bar. For pH measurements, a specialized UltraDeg® pH probe from Corr Instruments was used, designed to function at pressures up to 350 bar and temperatures reaching 305°C. A stirrer was incorporated to accelerate the achievement of thermodynamic equilibrium. Temperature was monitored using a thermocouple, while a pressure transducer was mounted on the upper section of the cell.

Throughout the experiments, carbon dioxide and brine were sequentially injected into the autoclave, with temperature and pressure adjusted to meet the required test conditions. The mixture was stirred for two hours and left to stabilize overnight. Once equilibrium was reached, a sample from the brine phase was extracted using a density meter and placed into a pre-weighed container. The CO_2-enriched brine underwent phase separation in the sampling cylinder and was then cooled. The mass of the sampling cylinder was measured, and the amount of dissolved CO_2 was determined by gradually releasing the gas until no further emission was observed. The cylinder was weighed again after CO_2 release. The overall volume of carbon dioxide dissolved in brine (*Dissolved* $CO2_{total}$) was calculated by Eq. 1.

$$Dissolved \quad CO2_{total}, \quad ml = CO2_g + CO2_r + CO2_d \qquad \text{Eq. 1}$$

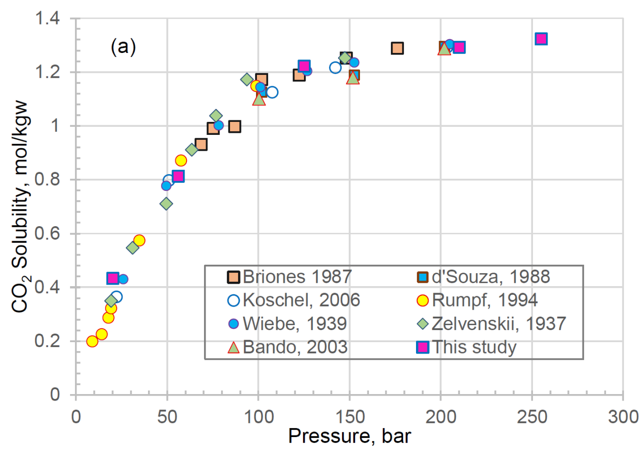
where CO_{2_g} denotes the gaseous CO_2 present inside the sampling cylinder, CO_{2_d} indicates the volume of CO_2 expelled from the sampling cylinder, and CO_{2_r} represents the residual dissolved CO_2 within the cylinder at ambient circumstances.

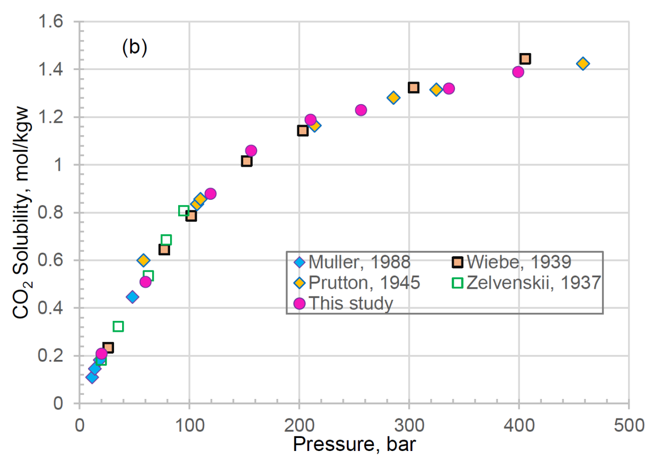
Phreeqc, an open-source software developed by the United States Geological Survey (USGS), is generally acknowledged as a benchmark for thermodynamics and geochemistry study. In Phreeqc, the CO_2 solubility in the aqueous phase was estimated by the Peng-Robinson EOS model and Henry's law based on gas fugacity and activity for mixed brines and gases (Peng, 1976). The geochemistry reactions can also be accurately modeled by Phreeqc. In this study, Phreeqc was coupled with MATLAB for CO_2 solubility and CO_2-brine-mineral interactions study, which is called CO_2SolTool in this study.

Results

CO_2-Water System

The experimental results measured by the in-house developed setup was first validated by comparing with the published data in the literature. The experimental method used in this work provided CO_2 solubility measurements in water that were in good agreement with literature data, as can be seen in Figures 2 (a) and (b), and hence, showed the merits of using this method for measuring CO_2 solubility in brine at high pressures (Briones, 1987; d'Souza, 1988; Koschel, 2006; Rumpf, 1994; Wiebe, 1939; Zelvenskii, 1937; Bando, 2003; Muller, 1988; Prutton, 1945). Subsequently, the simulation results by the CO_2SolTool were validated by experimental data of CO_2 solubility and properties of CO_2-saturated brine at different temperature, pressure and salinity.





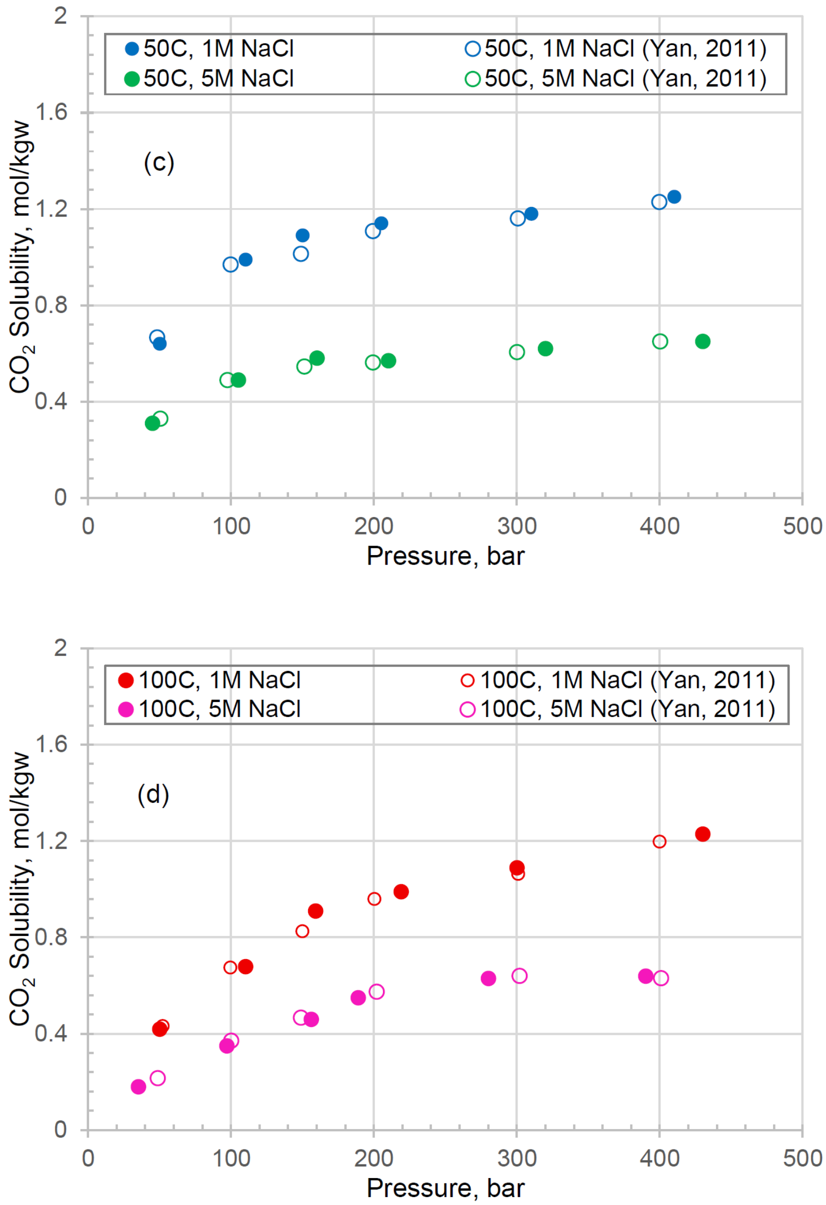


Figure 2—The CO_2 solubility in water (a) 50°C and (b) 100°C; the CO_2 solubility in 1M and 5M NaCl brine at (c) 50°C and (d) 100°C.

CO_2-NaCl Brine System

The saline aquifers typically contain a certain amount of salts, predominantly NaCl. It is, therefore, imperative to study the CO_2 solubility in brine. As shown in Figure 2 (c) and (d), the CO_2 solubility in NaCl brine measured by the setup is consistent with the published data (Wei, 2011). Moreover, the developed CO_2 SolTool can well reproduce the CO_2 solubility data under aquifer conditions that are technically realistic for carbon sequestration, as illustrated in Figure 3 (a). CO_2 solubility and the associated CO_2-saturated brine densities were measured from 50-400 bar, at 323 K, 373 K and 413 K, in 1 M and 5 M NaCl. The chemical reactions for CO_2 dissolution in the CO_2-brine system are elucidated in Eqs. 2 and 3.

$$CO_2(aq) + H_2O \underset{k1}{\leftrightarrow} H^+ + HCO_3^- \qquad \text{Eq. 2}$$

$$HCO_3^- \underset{k2}{\leftrightarrow} H^+ + CO_3^{2-} \qquad \text{Eq. 3}$$

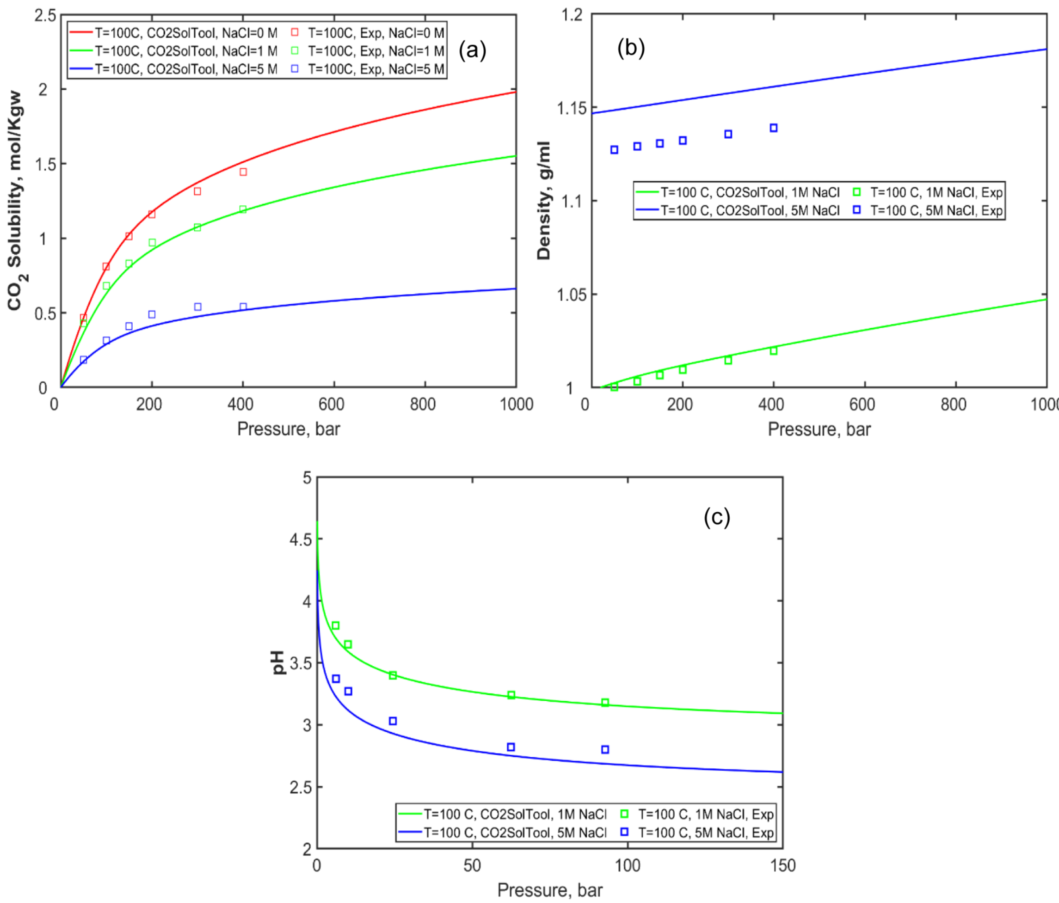


Figure 3—(a) CO_2 solubility in brine of different salinity under different pressures at 100°C, (b) density of CO_2 saturated brine, and (c) pH of CO_2 saturated brine at different salinity and pressure. The lines are calculated by CO_2SolTool, while the scatters are experimental data.

The density of CO_2-saturated brine at different temperatures, pressures, and salinity is elucidated in Figure 3 (b). The calculated data can reasonably match the experimental results (Haghi, 2017). The elevated brine density at elevated CO_2 pressure will result in density driven convection during CO_2 geological storage. An accurate input of the liquid density is also necessary. The pH of CO_2-saturated brine at different temperatures and pressure and NaCl molarity is illustrated in Figure 3 (c) (Haghi, 2017). The calculated results by CO_2SolTool can reasonably match the experimental data. Notably, the CO_2 solubility in high salinity brine is smaller, yet the pH is smaller compared to low salinity brine, which is probably attributed to ion interaction effects that increase the activity coefficients of CO_2 and H^+. Overall, the CO_2SolTool model performs well at various temperature and pressure ranges and different NaCl concentrations.

CO_2 solubility in mixed brine

The formation brine in standard saline aquifers is intrinsically a mixture of different brines, generally comprising various salts, hence complicating the accurate replication of its composition in laboratory experiments (Zhao, 2015). Studying the amount of CO_2 in synthetic brine with diverse electrolytes is equally crucial because of the existence of divalent ions in most saline aquifers. In this section, the effects of $CaCl_2$, $MgCl_2$, and other salts on CO_2 solubility are preliminarily studied (Zhao, 2015). The compositions of synthetic brine solutions for Mt. Simon (MS) and Michigan Basin (MB) are exhibited in Table 1.

Table 1—Synthetic Mt. Simon and Michigan basin brine compositions

	Mt. Simon formation brine (g/L)		Michigan Basin brine (g/L)	
	Synthetic	Proxy	Synthetic	Proxy
Sodium Chloride	61.95	61.95	174.48	174.48
Calcium Chloride	15.15	24.11	43.69	73.92
Sodium Sulfate	2.34	--	0.014	--
Magnesium Chloride	5.18	--	24.14	--
Potassium Chloride	1.40	--	1.66	--
TDS, ppm	84773	86125	238025	248595

The amount of dissolved CO_2 in brine for both the synthetic (MSS and MBS) and proxy brine (MSP and MBP) solutions presented in Table 1 is depicted in Figures 4 (a) and (b). It is demonstrated that the experimental data are in good agreement with the data calculated by CO_2SolTool. Therefore, the developed CO_2SolTool can be used to further study the CO_2 solubility at varying temperatures, pressures, and brine salinities and compositions. Divalent ions, such as Mg^{2+} and Ca^{2+}, can impact the CO_2 dissolution behavior.

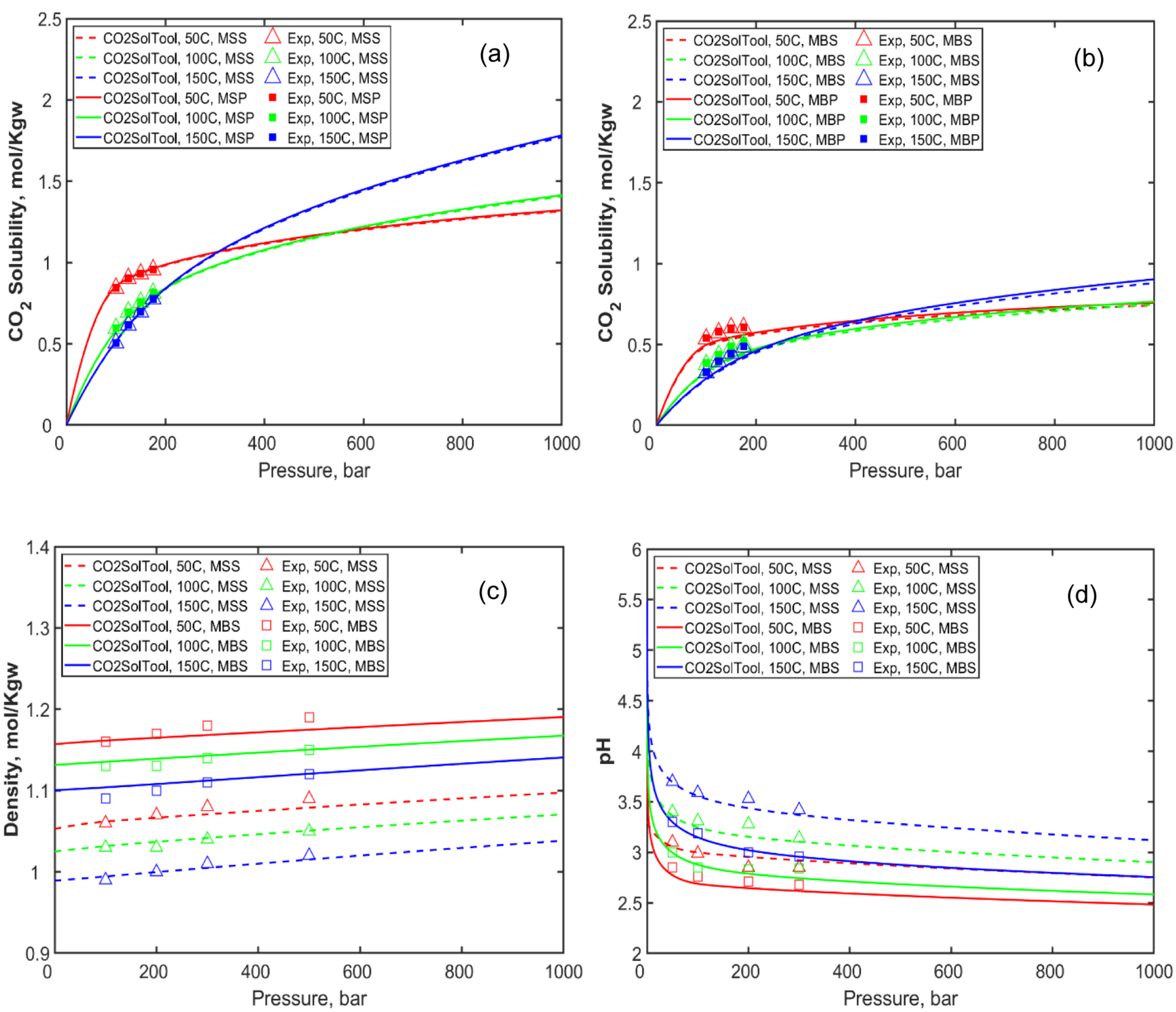


Figure 4—(a) Solubility of CO_2 synthetic Mt. Simon (MSS) and proxy Mt. Simon (MSP) brine. (b) Solubility of CO_2 in synthetic Michigan Basin (MBS) and proxy Michigan Basin (MBP) brine. (c) Density of CO_2 saturated brine and (d) pH of CO_2 saturated brine at varying salinity and pressure. The lines are calculated by CO_2SolTool, while the scatters are published experimental data.

Experiments were conducted in this study to determine the density and pH of synthetic MBS and MSS saline with high pressure of solubilized CO_2. Subsequently, the experimental results were compared with the calculated data. It is observed from Figures 4 (c) and (d) that the calculated density and pH of CO_2-saturated MSS and MBS brine align closely with the experimental results. The pH of CO_2-saturated MBS and MSS brine decreased as pressure and temperature increased, while the density of CO_2-saturated brine modestly increased with elevated pressure and decreased temperature. The CO_2SolTool serves as a robust tool in calculating the CO_2 solubility and properties of CO_2-saturated brine for multicomponent electrolytes brine solutions. During CO_2 geological storage, CO_2 can migrate upward, creating a CO_2 plume at the top of the aquifer. After CO_2 has been solubilized into the aqueous phase at high pressure, the liquid density may also moderately increase. The increase in brine density may induce a density-driven convection and enhance the CO_2 mixing and dissolution fingering (Bacon, 2014).

Impact of Gas Impurities

The majority of the CO_2 streams contain a considerable amount of other gases, such as H2S, N2, and CH4 (Wu, 2024; Savary, 2012; Mahmoodpour, 2020; Yu, 2021). Therefore, it is equally important to investigate the impact of these gas impurities on the properties of CO_2-saturated brine.

Influence of H_2S. Only limited data is available due to the challenges associated with performing experiments involving H_2S. The impact of H_2S on the solubility and pH of the CO_2-saturated water is summarized in Figure 5. Figure 5 (a) depicts the computed H_2S solubility in water and 1 M NaCl at 60°C. The experimental data is from the literature (Søreide, 1992; Charlton, 2011; Li, 2018). It is found that the H_2S solubility is higher than that of CO_2 under identical testing conditions. Moreover, the predicted H_2S solubility is consistent with those documented in the literature. At a constant CO_2 pressure, the existence of H_2S will diminish the amount of CO_2 dissolved in brine, as illustrated in Figure 5 (b).

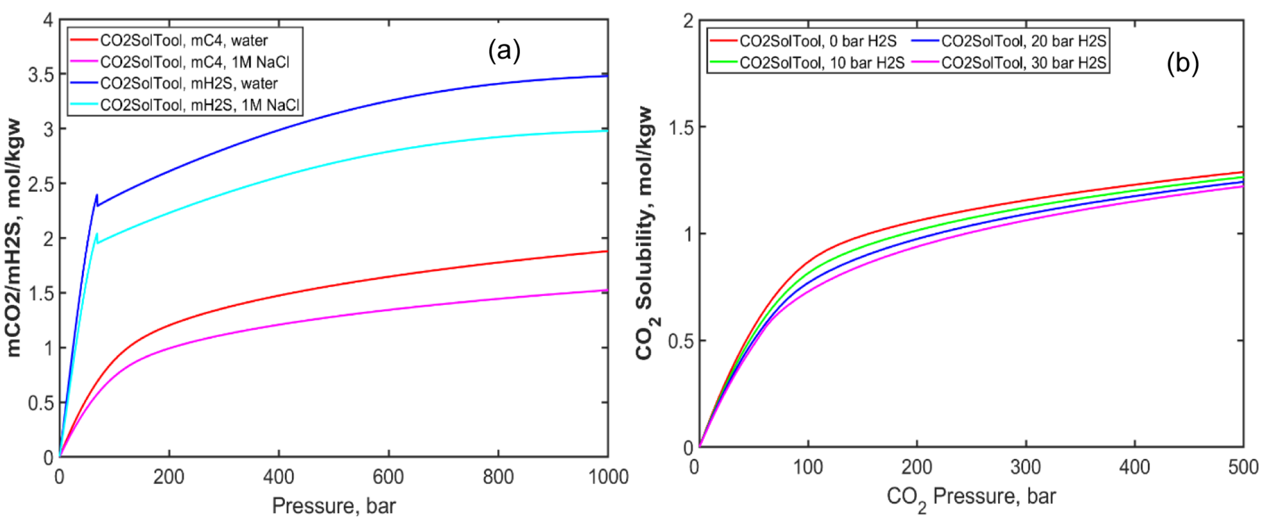


Figure 5—(a) solubility of pure CO_2 and pure H2S in water and 1M NaCl, (b) The influence of H2S on CO_2 solubility in water under different pressure

Influence of N_2. Figure 6 (a) illustrates the calculated N_2 solubility in water and 4 M NaCl at 60°C. The solubility of N_2 in brine is minimal in comparison to CO_2. The presence of N_2 has a moderate effect on CO_2 solubility at elevated CO_2 pressure (fugacity). Increasing the pressure of N_2 may moderately decrease the fugacity and solubility of CO_2 at a constant pressure. At a constant pressure of CO_2, the presence of N_2 will slightly diminish the CO_2 solubility in brine, as illustrated in Figure 6 (b) (Duan, 1992; Duan, 2007; Foltran, 2015; Nakhaei-Kohani, 2022).

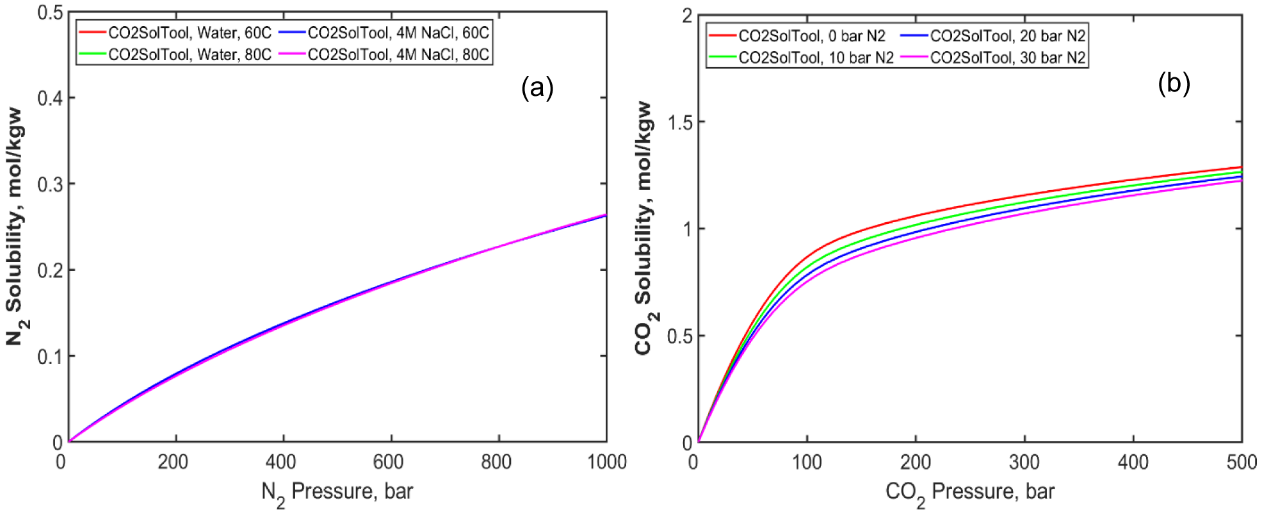


Figure 6—(a) Solubility of N2 in water and 4M NaCl, (b) The influence of N2 on dissolved CO_2 in water at different pressure.

Influence of CH_4. Methane (CH_4) is the other primary source of greenhouse gas (Duan, 1992; Zirrahi, 2012). CH_4 is also relatively inertial, and its impact on CO_2 solubility and properties of CO_2-saturated brine is illustrated in Figure 7. Figure 7 (a) demonstrates that the solubility of CH_4 in water is analogous to that of N_2, both exhibiting minimal sensitivity to the brine salinity. When the pressure of CH_4 is increased, the fugacity and solubility of CO_2 at a fixed pressure may be moderately decreased (Figure 7(b)).

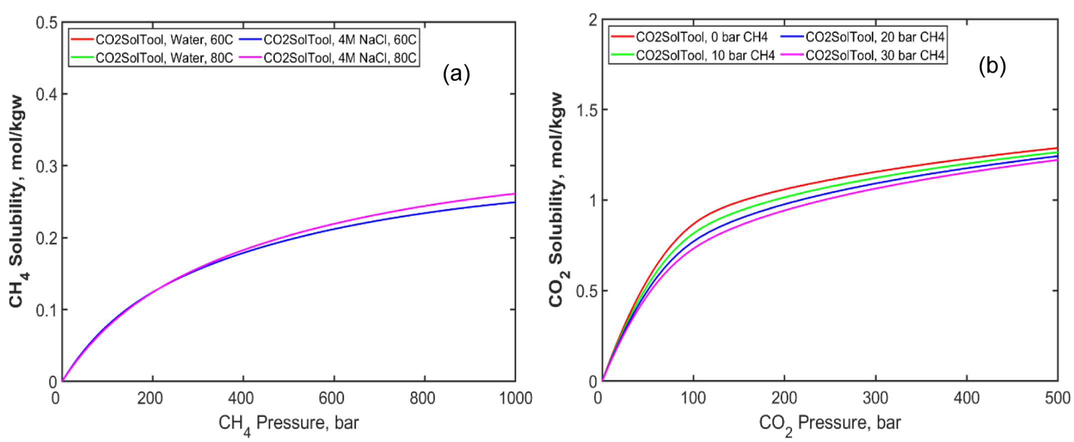


Figure 7—(a) Solubility of CH4 in water and 4M NaCl, (b) The influence of CH4 on pH of CO_2 saturated brine at different pressure.

CO_2-Brine-Mineral Interactions

Reactive minerals can interact with dissolved CO_2, influencing its overall concentration and distribution in the brine (Li, 2018; Kharaka, 2009; Jeon, 2018). Dissolved CO_2 can generate carbonic acid in brine, which dissociates into bicarbonate and carbonate ions (Wei, 2011). The presence of minerals, including calcite, may significantly affect the geochemistry interactions between CO_2, brine, and mineral, as depicted in Figure 8. Other compositions, such as Ca^{2+}, HCO_3^-, and CO_3^{2-}, will also be altered in the presence of calcite. As indicated by Figures 8 (a) and (b), the presence of calcite may slightly reduce the amount of free CO_2 solubilized in water. It is found that the amount of C4 (i.e., the total amount of free CO_2 in brine, HCO_3^-, and CO_3^{2-}) is increased in the presence of calcite, due to the enhanced CO_2 dissociation and mineral dissolution, as exhibited by the red lines in Figure 8 (b). A certain amount of C4 is originated from calcite dissolution, and this quantity would be equal to the amount of Ca^{2+} (Figure 8 (c)). The contribution from calcite dissolution can subsequently be quantitatively assessed. It is found that the chemical equilibrium may be altered, leading to a modest increase in the amount of dissociated CO_2 in the presence of calcite, as depicted by the cyan and blue lines in Figure 8 (b).

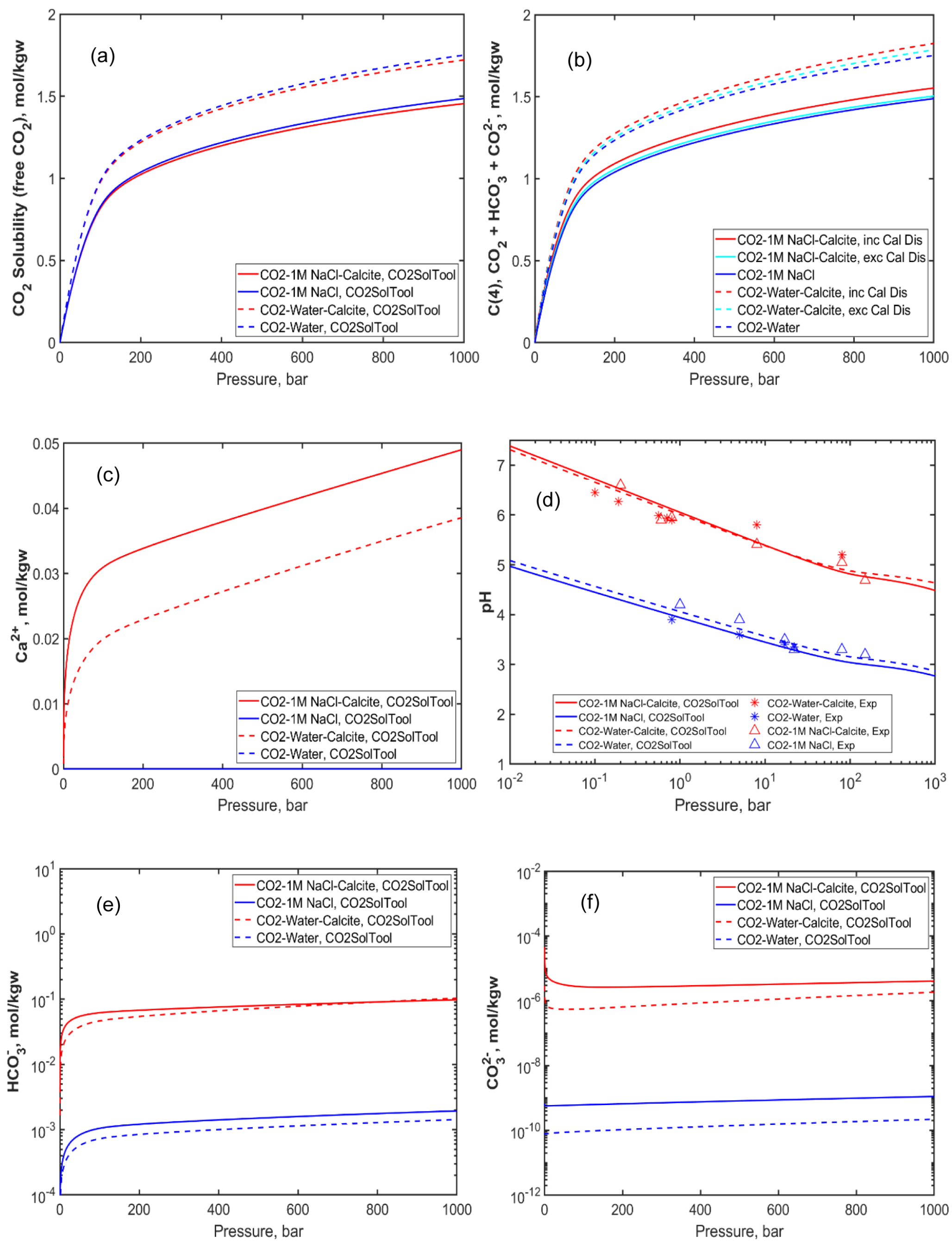


Figure 8—(a) The dissolved free CO_2, (b) the dissolved C4, (c) the pH of CO_2-saturated water, (c) the Ca2+ concentration, (d) the pH, (e), the HCO3- concentration, and (f) the CO32- concentration in pure water and 1M NaCl at 65°C with and without calcite.

The existence of calcite generally increases the pH of CO_2-saturated water and brine, as indicated by Figure 8 (d). The amount of Ca^{2+} signifies that the calcite dissolution is more pronounced in high salinity brine due to the reduction of pH at the elevated salinity, as demonstrated in Figures 8 (c) and (d). Figures 8 (e) and (f) also demonstrate that the compositions of HCO_3^- and CO_3^{2-} would also be slightly altered in the presence of calcite. Exposure to differing CO_2 pressures results in variations in the pH and ionic content of CO_2-saturated brine (Zhang, 2024). The chemical reactions for mineral dissolution and mineralization reactions the CO_2-water-calcite system are elucidated in Eqs. 4 and 5. The pH of the brine influences CO_2 solubility through equilibrium reactions that convert CO_2 into bicarbonate and carbonate ions.

$$CaCO_3 + \quad H^+ \underset{k3}{\longleftrightarrow} \; Ca^{2+} + HCO_3^- \qquad \text{Eq. 4}$$

$$CaCO_3 + \quad 2H^+ \underset{k4}{\longleftrightarrow} \; Mg^{2+} + Ca^{2+} + 2HCO_3^- \qquad \text{Eq. 5}$$

Influence of Carbonate Potential Determining Ions

Influence on CO_2 solubility and pH. The PDI, namely HCO_3^- and CO_3^{2-} for carbonates, significantly affects the wettability and surface charge of carbonate minerals (Daryasafar, 2019; Sun, 2021). Studying the influence of carbonate PDI on the quantity of CO_2 that can be dissolved in brine is also crucial. The results are illustrated in Figure 9 (a) and (b) (Han, 2011). The addition of HCO_3^- ions can also change the brine compositions at equilibrium, according to Eqs 2-5. It is concluded that the presence of bicarbonate ions can slightly decrease the free CO_2 solubility in brine, while concurrently increasing the quantity of dissociated CO_2. The pH of the bicarbonate solution after dissolving CO_2 is illustrated in Figure 9 (c) (Han, 2011; Li, 2018). The CO_2SolTool is capable of accurately predicting the amount of dissolved CO_2 in bicarbonate solution and the pH of the CO_2-saturated brine solution. The addition of HCO3- has a buffering effect, making the pH drop in the bicarbonate solution that is saturated with CO_2 less noticeable than that without the bicarbonate. The brine compositions, pH, and ionic strength affect these equilibria and therefore the solubility of CO_2.

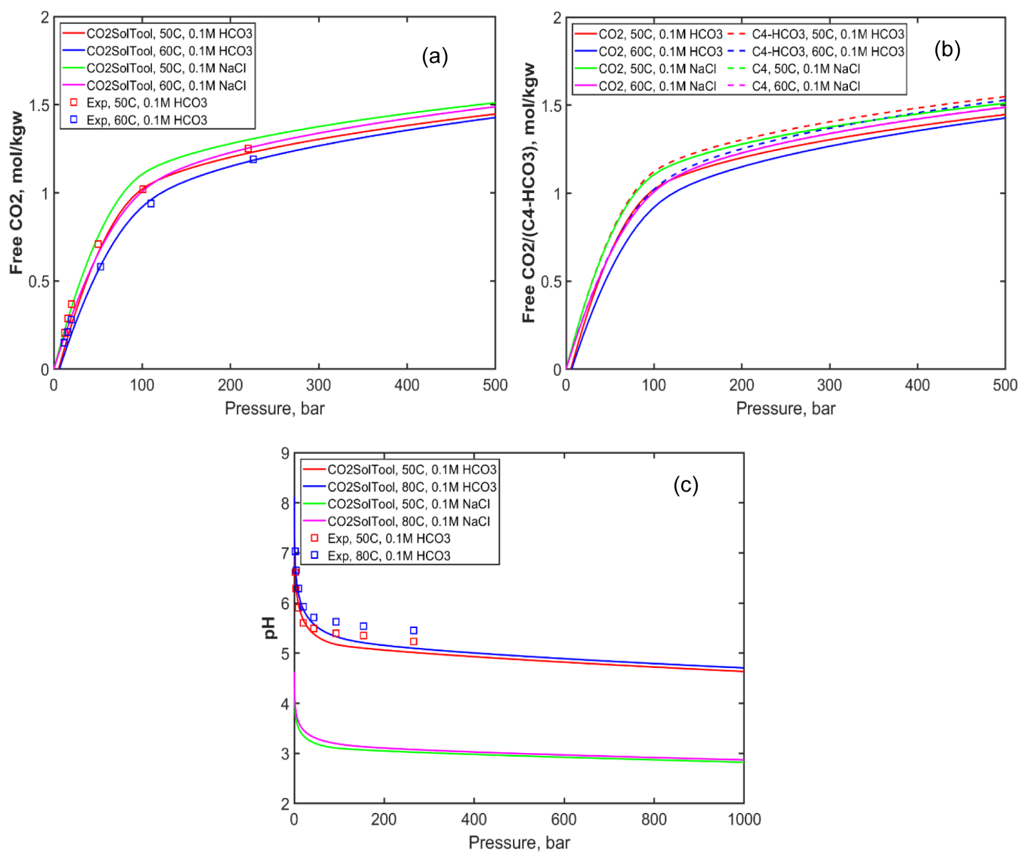


Figure 9—(a) Solubility of CO_2 in sodium bicarbonate brine, (b) the free CO_2 and C4 (excluding added 0.1 m HCO_3^-) concentrations in brine, and (c) the pH of CO_2 in sodium bicarbonate brine. The lines are calculated by CO_2SolTool, while the scatters are published experimental data.

Influence on surface charge. The brine compositions, specifically the PDIs, would significantly impact the wettability and surface charge of minerals. The bicarbonate ion is the PDIs for calcite (Cui, 2015; Ma, 2013). The influence of bicarbonate ions on surface charge of calcite is illustrated in Figure 10 (a). The addition of HCO_3^- tends to decrease the surface potential, while the addition of Ca^{2+} may increase the electrical charge of the surface of calcite in the calcite-brine-CO_2 system.

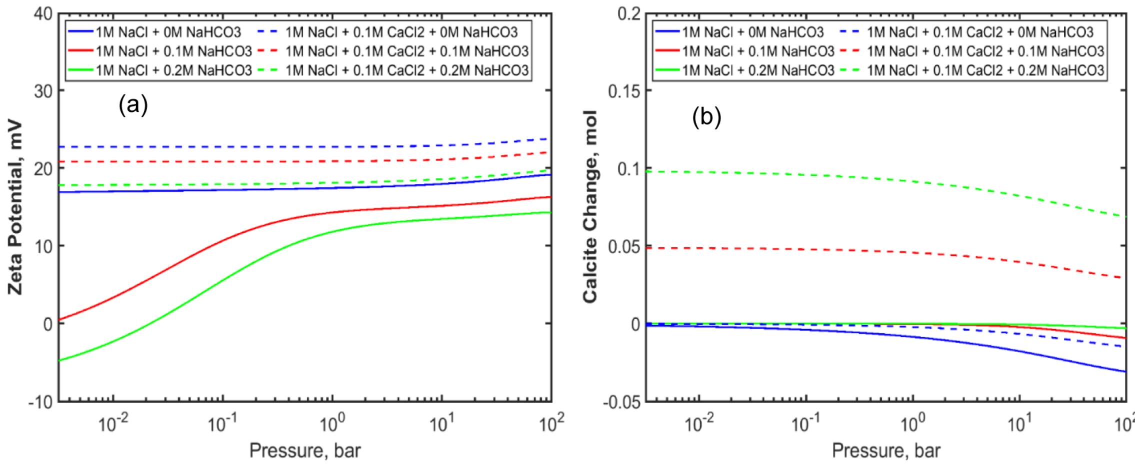


Figure 10—(a) The influence of bicarbonate on zeta potential of calcite in the calcite-brine-CO_2 system, and (b) the influence of bicarbonate on dissolution/precipitation of calcite in the calcite-brine-CO_2 system

Conclusions

CO_2 solubility trapping and CO_2-brine-mineral interactions are essential in the long-term reliability of geological storage of CO_2. Primarily, a high-fidelity setup was developed to measure the CO_2 solubility in brine and evaluate the CO_2-brine-mineral interactions. The experimental results were validated by published data in the literature. However, measuring these properties are time consuming and tedious. Coupling MATLAB with Phreeqc provides an alternative approach for accurately calculating the CO_2 solubility, properties of CO_2-saturated brine, and CO_2-brine-mineral interactions. The calculated data for CO_2 solubility, CO_2-brine-mineral interactions, and properties of CO_2-saturated brine (density and pH) are consistent with the experimental data across a wide range of temperature, pressure, and brine salinity.

The solubility of H2S is higher than that of CO_2, while the solubility of N_2 and CH_4 is smaller than that of CO_2. The electrolytes, especially the HCO_3^-, have a significant effect on the pH of CO_2-saturated brine. The presence of impurities such as H_2S may moderately decrease the pH of the CO_2-saturated brine, as it has higher solubility in brine than CO_2 under the same conditions. The presence of minerals, e.g., calcite and dolomite, may increase the pH level after CO_2-brine-mineral equilibrium. The pH of CO_2-saturated brine is not significantly affected by the presence of anhydrite and gypsum. The zeta potential of calcite is largely influenced by the potential-determining ions, HCO_3^-, CO_3^{2-}, Ca^{2+}, etc. Predicting CO_2 solubility and CO_2-brine-mineral interactions allows for better modeling of the fate and transport of CO_2 within the subsurface, optimizing storage strategies, and predicting the long-term behavior of CO_2 in geological formations.

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